

SAS Advanced Analytics Healthcare Project

U.S. Hospital Patient Experience & Service Issue Analysis

Dataset: U.S. Hospital Customer Satisfaction 2016–2020 (Kaggle / CMS HCAHPS)

Download link: <https://www.kaggle.com/datasets/abrambeyer/us-hospital-customer-satisfaction-20162020>

Project objective: Analyze patient-experience measures reported for U.S. hospitals, identify major service issues, compare hospitals and states, study trends, segment hospitals, and build predictive models for poor patient experience.

1. Dataset Understanding – Questions

1. What is the source of the dataset?
2. What years are covered by the dataset?
3. How many rows and columns are available in each annual file?
4. What does one row represent?
5. What is HCAHPS?
6. What are the main patient-experience measures?
7. Which variables identify the hospital?
8. Which variables describe the location of the hospital?
9. Which variables measure patient experience?
10. Which variables contain ratings, percentages, or scores?

2. SAS Data Import & Data Quality

11. Import the 2016–2020 CSV files into SAS.
12. Display the first 10 observations.
13. Use PROC CONTENTS to examine variable names and data types.
14. Count the number of observations in each yearly dataset.
15. Check for missing values in every important variable.
16. Identify variables containing 'Not Available', 'Not Applicable', or footnote values.
17. Check duplicate hospital/measure records.
18. Check whether numeric percentage fields were imported as character variables.

19. Convert required character variables into numeric variables.
20. Create one combined SAS dataset for all years.
21. Create a YEAR variable.
22. Document all data-cleaning decisions.

3. Exploratory Data Analysis

23. What is the overall distribution of patient-experience scores?
24. What are the most frequently reported HCAHPS measures?
25. Which patient-experience measures have the highest average scores?
26. Which measures have the lowest average scores?
27. What percentage of hospitals have high patient-experience ratings?
28. Which states have the highest average patient-experience scores?
29. Which states have the lowest average scores?
30. Which cities/counties contain the highest-performing hospitals?
31. Which hospitals have consistently strong patient-experience results?
32. Which hospitals have consistently weak results?

4. Patient Complaint / Service Issue Analysis

33. What are the major patient-experience problem areas?
34. Is communication with nurses a major issue?
35. Is communication with doctors a major issue?
36. Is staff responsiveness a major issue?
37. Is communication about medicines a major issue?
38. Is discharge information a major issue?
39. Is hospital cleanliness a major issue?
40. Is quietness/rest a major issue?
41. Which issue has the largest performance gap?
42. Which service issue should hospital management prioritize first?

- 43. Create an overall Patient Experience Issue Score.
- 44. Classify hospitals into Low, Medium, High, and Critical issue groups.

5. Time Trend Analysis – 2016 to 2020

- 45. How did patient experience change from 2016 to 2020?
- 46. Which measures improved the most?
- 47. Which measures deteriorated?
- 48. Which states improved the most?
- 49. Which states deteriorated the most?
- 50. Which hospitals show sustained improvement?
- 51. Which hospitals show sustained deterioration?
- 52. Did patient experience change significantly over time?

6. Statistical Analysis

- 53. Does patient experience differ significantly by state?
- 54. Does patient experience differ significantly by hospital group?
- 55. Is overall hospital rating associated with patient-experience measures?
- 56. Which patient-experience variables have significant relationships with overall rating?
- 57. Perform appropriate Chi-square tests for categorical variables.
- 58. Perform t-tests where two groups are compared.
- 59. Perform ANOVA when comparing multiple groups.
- 60. Perform Tukey post-hoc analysis where appropriate.
- 61. Build a regression model to explain overall patient-experience performance.

7. Hospital Segmentation / Clustering

- 62. Can hospitals be grouped according to their patient-experience profiles?
- 63. Create hospital-level summary variables.
- 64. Standardize variables before clustering.
- 65. Determine a suitable number of clusters.
- 66. Build a clustering solution.

- 67. Describe each hospital segment.
- 68. Identify the high-performing cluster.
- 69. Identify the average-performing cluster.
- 70. Identify the high-risk/poor-experience cluster.

8. Predictive Analytics

- 71. Define a target variable for poor patient experience.
- 72. What variables should be used as predictors?
- 73. Build a Logistic Regression model.
- 74. Interpret significant predictors and odds ratios.
- 75. Build a Decision Tree model.
- 76. Build a Random Forest model.
- 77. If available in the SAS environment, build a Gradient Boosting model.
- 78. Compare all models.
- 79. Calculate ROC-AUC.
- 80. Calculate accuracy, precision, recall, specificity, and F1 score.
- 81. Produce confusion matrices.
- 82. Determine the best predictive model.
- 83. Identify the most important variables.

9. Model Validation & Risk Scoring

- 84. Split data into training, validation, and test samples.
- 85. Check whether class imbalance affects model performance.
- 86. Validate the selected model on unseen data.
- 87. Create predicted probabilities.
- 88. Create hospital risk categories.
- 89. Identify high-risk hospitals.
- 90. Produce a ranked list of hospitals/service areas requiring attention.

91. Evaluate lift and gains.

92. Determine how many hospitals could be prioritized for intervention.

10. Business Interpretation

93. What are the top five patient-experience problems?

94. Which states need the most improvement?

95. Which hospitals are consistently high performers?

96. Which hospitals are consistently poor performers?

97. What are the strongest predictors of poor patient experience?

98. Which interventions should hospital management prioritize?

99. How can hospitals use the model for continuous monitoring?

100. What are the limitations of the analysis?

11. Final Deliverables

101. Prepare an executive summary.

102. Prepare a SAS analytical report.

103. Create charts for major findings.

104. Create a state-level performance table.

105. Create a hospital-level risk table.

106. Create a patient-experience issue matrix.

107. Create a model comparison table.

108. Create final management recommendations.

109. Document assumptions and limitations.

110. Prepare a 5–10 minute presentation of the project.

Suggested Project Title

U.S. Healthcare Patient Experience Analytics: Identifying, Predicting and Prioritizing Hospital Service Issues Using SAS

Suggested Folder Structure

01_Data

02_Import

03_Data_Quality
04_EDA
05_Statistics
06_Feature_Engineering
07_Clustering
08_Predictive_Models
09_Model_Validation
10_Risk_Scoring
11_Report
12_Presentation

Submission Guidelines

- Last Date of submission: **18th Oct 2026**
- Submit all project files in a single compressed (.zip) folder.
- Name the folder using the format: **YourName_Projecttitle.zip**.
- Ensure that all SAS programs execute without errors.
- Clearly comment your code to improve readability and maintainability.
- Any assumptions made during data cleaning or analysis should be documented in the report.
- Late submissions may be subject to penalties as per the course policy.